Quick tip: Review the prerequisites before you run the lab

End Lab 00:25:35

Caution: When you are in the console, do not deviate from the lab instructions. Doing so may cause your account to be blocked. Learn more.

Open Google Cloud console

Username

student-02-9f81efeeda67(

Password

EJrnaIvhic93

GCP Project ID

qwiklabs-gcp-02-d457d8f!

Search and Conversational AI in Vertex AI and Conversational Agents: Challenge Lab

Lab · 1 hour · No cost · Introductory

This lab may incorporate AI tools to support your learning.

Previous

Next

The objective of this challenge lab is for you to demonstrate proficiency in integrating Search in applications using AI Applications, showcasing your ability to design and implement a comprehensive solution that combines powerful search capabilities and a virtual assistant to enhance the user experience.

Setup

Qwiklabs setup

For each lab, you get a new Google Cloud project and set of resources for a fixed time at no cost.

1. Make sure you signed into Qwiklabs using an **incognito window**.
2. Note the lab's access time (for example, 02:00:00 and make sure you can

There is no pause feature. You can restart if needed, but you have to start at the beginning.

3. When ready, click START LAB .
4. Note your lab credentials. You will use them to sign in to the Google Cloud Console.

Open Google Console

Caution: When you are in the console, do not deviate from the lab instructions. Doing so may cause your account to be blocked. Learn more.

Username

google2876526_student@qwiklabs.n

Password

GCP Project ID

qwiklabs-gcp-0855e773352d3560

New to labs? View our introductory video!

5. Click **Open Google Console**.
6. Click **Use another account** and copy/paste credentials for **this** lab into the prompts.

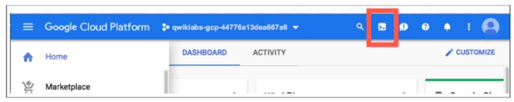
If you use other credentials, you'll get errors or **incur charges**.

7. Accept the terms and skip the recovery resource page.

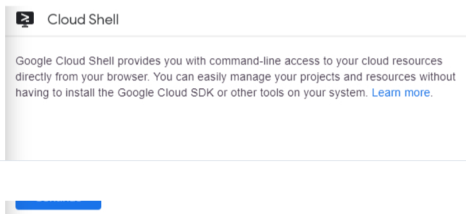
Do not click **End Lab** unless you are finished with the lab or want to restart it. This clears your work and removes the project.

Cloud Shell is a virtual machine that is loaded with development tools. It offers a persistent 5GB home directory and runs on the Google Cloud. Cloud Shell provides command-line access to your Google Cloud resources.

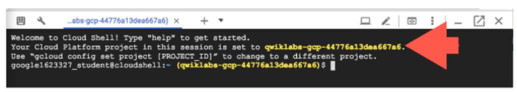
In the Cloud Console, in the top right toolbar, click the **Activate Cloud Shell** button.



Click **Continue**.



It takes a few moments to provision and connect to the environment. When you are connected, you are already authenticated, and the project is set to your *PROJECT_ID*. For example:



`gcloud` is the command-line tool for Google Cloud. It comes pre-installed on Cloud Shell and supports tab-completion.

You can list the active account name with this command:

```
gcloud auth list
```

(Output)

```
Credentialed accounts:
```

(Example output)

```
Credentialed accounts:
 - google1623327_student@qwiklabs.net
```

You can list the project ID with this command:

```
gcloud config list project
```

(Output)

```
[core]
project = <project_ID>
```

(Example output)

```
[core]
project = qwiklabs-gcp-44776a13dea667a6
```

For full documentation of `gcloud` see the gcloud command-line tool overview.

Challenge Scenario

Here is a company overview as provided on the Cymbal Shops's website.

> *Cymbal Shops is an American retail chain headquartered in Minneapolis that sells*
>
> *Founded in 1974, Cymbal Shops started out as Cymbal Air, selling AC systems manufactured by Cymbal Group in Minnesota and neighboring states. The company quickly expanded into domestic merchandise in order to satisfy the need for quality products at an affordable price in the midst of a recession.*
>
> *Cymbal Shops' broad product assortment was once a benefit - capturing planned purchases and last minute splurges. In recent years, however, the company has struggled to adapt to the acceleration in e-commerce. Digitally native companies are on the rise and Cymbal Shops must implement significant changes in order to keep pace and maintain relevance. Today, Cymbal Shops operates 714 stores across North America and reported $15 billion in revenue in 2019. They currently employ 80,400 employees across the United States and Canada.*
>
> *Cymbal Shops is a digitally transforming legacy retailer.*
>
> *It is inspired by clients like: Bed Bath & Beyond, Best Buy, Home Depot, Nordstrom.*

You work as an engineer at Cymbal Shops, a retail chain company renowned for its innovative solutions. Cymbal Shops is embarking on a significant project that involves developing a robust search functionality and virtual agent for a new application. Your task is to design a Proof of Concept (POC) solution to understand the implications and demonstrate the capabilities of integrating AI Applications and Data Store Agent in the application.

Cymbal Shops' new application demands a powerful and efficient search feature, aiming to catalog and retrieve various data types, including text documents, images, and structured data using AI Applications Search. Additionally, the company aims to create a virtual agent that can assist customers with questions about products and devices available in the Google Store, such as phones, watches, laptops, smart home devices, and other consumer devices. This virtual agent will be built and configured using AI Applications Conversation.

Your objective is to craft a POC solution that effectively showcases the capabilities of both AI Applications Search and Data Store Agent within the context of this application, enabling powerful search capabilities and a virtual assistant to enhance the customer experience.

Therefore in this lab, you should:

2. Create a structured data search app `cymstruc`.
3. Create an unstructured data search app `cymunstruc`.
4. Create a generative chat application named `cymbalsearch`.
5. Enhance customer engagement and optimize performance.

Task 1. Create a website search app

In this task, demonstrate how AI Applications Search can be utilized to create a web search application, showcasing its effectiveness in searching and retrieving content from websites. This app should be capable of crawling web content, indexing it, and providing users with the ability to search for information.

Therefore in this task:

`Global`.

2. Configure this app to use a datastore named `cymwebstore` that indexes data from the following website: cloud.google.com/*.
3. Finally, preview the app and verify if it is working as intended.

Click **Check my progress** to verify the objectives.

Create a website search app

Check my progress

Assessment Completed!

Task 2. Create a structured data search app

searching structured information. The application should allow users to search for specific data points, filter results, and navigate through structured information efficiently.

Therefore in this task:

1. Create an app named `cymstruc` of type `Search` and set the location of the app to `Global`.

2. Configure this app to use a datastore named `cymbalstore` that ingests data from the file present in the following Cloud Storage bucket: `cloud-samples-data/gen-app-builder/search/kaggle_movies/movie_metadata.ndjson`.

Hint: This Cloud Storage bucket contains Newline Delimited JSON-formatted dataset of movies made available by [Kaggle](#).

3. Preview the `cymbalstore` app and configure the search results to only display the following:

Key	Value
Title	title
Text 1	tagline
Text 2	revenue

4. Finally, preview the app and verify if it is working as intended.

Click **Check my progress** to verify the objectives.

Create a structured data search app

Check my progress

Assessment Completed!

Task 3. Create and preview an unstructured data search app

unstructured data search application, illustrating its proficiency in searching and presenting unstructured content.

Therefore, in this task,

1. Create an app named `cymunstruc` of type `Search` and set the location of the app to `Global`.

2. Configure this app to use a datastore named `cymwarehouse` that ingests unstructured data from the following Cloud Storage bucket: `cloud-samples-data/gen-app-builder/search/alphabet-investor-pdfs`.

3. Finally, preview the app and verify if it is working as intended.

Click **Check my progress** to verify the objectives.

Create a unstructured data search app

Check my progress

Assessment Completed!

Task 4. Create a generative chat application

In this task, you should create a Data Store Agent using AI Applications Conversation. This agent should be able to comprehend unstructured data. Your objective is to configure and deploy a virtual agent that can provide assistance to customers, with a particular focus on handling unstructured information.

Therefore, in this task,

1. Enable the **Dialogflow** and **AI Applications** APIs.

2. Navigate to [Conversational Agents](#), select your project and create a `Conversational Agents` application with the existing agent named

conversational Agents application with the existing agent named cymbalsearch.

3. Configure this application to use a `cymappstore` that ingests unstructured data from the folder present in the following Cloud Storage bucket: `cloud-samples-data/gen-app-builder/search/alphabet-investor-pdfs`.

4. Navigate to **Conversational Agents > Flows**, in the `Start Page`, add a `Data Store` state handler to use your `cymappstore` data store.

5. Test the virtual agent by starting an interactive session with the chatbot to see how it responds to various questions that a customer might ask.

   a. Type a greeting to your agent such as `Hello`.

   b. Ask the agent some questions such as.

```
i. What was the revenue for Google in 2004?

ii. Where does Google see the most potential opportunity
for long term growth?

iii. Where is Google allocating the most amount of its
capital?
```

Note: If your agent is giving you responses from the default negative intent (e.g., "I'm sorry, I didn't get that. Can you rephrase your question?"), then be aware that it can

Click **Check my progress** to verify the objectives.

Create a generative chat application

Check my progress

Assessment Completed!

Task 5. Enhance customer engagement and optimize performance

In this task, you will set up both voice and chat interactions for your virtual agent, created with AI Applications Conversation. The objective is to create versatile communication channels that allow customers to engage with your agent through voice commands and text-based chat, delivering a flexible and user-friendly experience. You

view and interpret data related to customer interactions, agent responses, and overall performance. Understanding these analytics is essential for optimizing your virtual agent's capabilities and enhancing customer service.

Therefore in this task;

1. Integrate a phone gateway into your bot named `cymbalsearch` for the United States. Name this phone gateway as `cymbalvoice`.

Note: This bot makes use of the Speech-to-Text and Text-to-Speech capabilities in Google Cloud.

2. Embed a chat widget on a website so customers can chat with it in addition to making a phone call to speak with it.

3. Finally, enable interaction logging for conversation history and analytics to view statistics related to agent requests and responses, and determine where the problems and friction points are in your virtual agent.

Click **Check my progress** to verify the objectives.

Enhance customer engagement and optimize performance

Assessment Completed!

Congratulations!

You've successfully integrated Search in applications with AI Applications.

Your points will be automatically totalled up in the system. You may end this lab now and run it again to try to finish faster which will earn more points in the competition!

Manual Last Updated April 03, 2025

Lab Last Tested April 03, 2025

Agents: Challenge Lab